David Nkpa

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Profile

Computer Science graduate (First-Class Honours, University of Leicester, 2026) who builds where AI, finance and data meet. Won 1st place at the UKFinnovator London Hackathon 2026 with a market-regime agent, built a cost model and executive deck with Rolls-Royce, and shipped analytics features at a London fintech. I care most about outputs people can check before they rely on them.

Work Experience

Finance Analyst Apr 2026 – Jun 2026
Rolls-Royce *London, UK*

- Consolidated a 12-month cost dataset (6 months of actuals and 6 of forecasts) into clean analysis for management review.
- Built an Excel model calculating monthly % variances, cumulative spend movement, run-rate trends and forecast accuracy across departmental cost lines.
- Turned the analysis into a 5-slide executive deck, with charts and plain-language commentary supporting 3 recommended actions.

Software Engineering Intern Sep 2025 – Oct 2025
Boom Technologies (Fintech) *London, UK*

- Built 10+ admin analytics features in a React back-office system, giving the team clearer operational data and faster decisions.
- Wrote an approved CEO-ready strategy report on LLM options and sustainable data-centre architectures, covering model trade-offs, data governance and deployment constraints for non-technical leadership.
- Documented proposed AI solutions and worked across teams to clarify requirements and ship features to deadline.

Research Intern Jun 2025 – Jul 2025
181st Street Communications *London, UK (remote)*

- Built website prototypes around the user journey and applied digital analytics and user research to onboarding flows, lifting sign-up conversion by 12% on test pages.
- Turned qualitative and quantitative findings into clear recommendations and test plans for non-technical stakeholders.

Strategy Consultant Intern Jun 2024 – Aug 2024
S&G Strategy *Montreal, Canada (remote)*

- Researched 40+ transport providers to assess the market, pricing models and competitive position, comparing leverage and liquidity indicators with operating performance.
- Condensed 350+ pages of material into a client-ready 50-slide presentation and presented recommendations to senior consultants.

Information Technology Intern Jun 2022 – Aug 2022
Ecobank Transnational Incorporated *Lomé, Togo*

- Supported critical banking systems (SWIFT, AML tooling) in a regulated environment, resolving 30+ priority network incidents under strict escalation and audit-trail requirements.
- Traced failures through logs across network and infrastructure layers and documented fixes for reuse.

Projects

Agenclave: Verification-Backed Trust Layer for Multi-Agent Coding Jul 2026
Python, FastAPI, React, scikit-learn, NumPy, Docker

- Built and shipped an MVP that runs each AI-generated code patch instead of only reading it, ranking candidates as trusted, applies-but-fails or broken with a plain-English reason; reviewed by a BLACKBOXAI co-founder.
- Trained a triage classifier (TF-IDF + logistic regression) reaching 0.675 macro-F1 on 1,793 held-out issues; kept it after measuring that it beat a heavier transformer-embedding approach.
- Built a Thompson-sampling router that learns each model's reliability from verified runs only.

Live DeFi Liquidity Agent Jun 2026
Python, Hidden Markov Models, NumPy

- Won 1st place at UKFinnovator London Hackathon 2026, Imperial College London; pitched to judges and challenge setters from King's College London and TurinTech AI.

- Built a Hidden Markov Model agent in pure NumPy that detects three market regimes in real time to manage concentrated liquidity in a Uniswap V3-style AMM simulation.
- Rebalanced only when expected profit beat the gas cost; outperformed all provided baselines on hidden evaluation seeds.

AI Financial Decision Assistant (Final-Year Project) Sep 2025 – May 2026

Python, FastAPI, React, PostgreSQL, RAG, LLMs

- Full-stack web app that helps retail investors read US stocks through volatility, drawdown and risk/return.
- Built a Python/FastAPI service that ingests Twelve Data market data and computes indicators with Pandas/NumPy across a 55-stock universe grouped into four K-Means risk clusters.
- Cut LLM hallucination by grounding every explanation in two-stage retrieval (risk band, then stock cluster), with hard rules against buy/sell calls and predictions.

Metropolitan Police Crime Prediction Model Mar 2026

Python, Pandas, scikit-learn, Random Forest, K-Means, MapReduce

- Built a pipeline that cleans 1.1M+ Metropolitan Police crime records and aggregates them into 56,571 neighbourhood-month rows.
- Engineered lagged counts, crime-type ratios and Shannon entropy features, with leakage checks and time-based splits.
- Reached 0.934 test R^2 with Random Forest; previous-month crime count was the strongest predictor.

Private Cloud Architecture on AWS Nov 2025 – Dec 2025

AWS, EKS, Kubernetes, ElastiCache (Redis), VPC

- Designed a fault-tolerant EKS architecture with horizontal scaling and self-healing workloads, isolated with VPCs, public/private subnets and NAT gateways.
- Ran shared state on ElastiCache (Redis), consistency-first under network partitions after reasoning through the CAP trade-offs.

Education

University of Leicester Leicester, UK

BSc Computer Science, First-Class Honours *Sep 2023 – Jun 2026*

- Citizens of Change 100 Scholarship (2023).
- **Relevant modules:** Big Data and Predictive Analytics; Data Analytics; Databases and Domain Modelling; Computers, Society and Professionalism.

The British School of Lomé Lomé, Togo

IGCSE (8 subjects: 7 A and 1 A, including Maths, ICT and Physics)* *Aug 2017 – Jun 2021*

Certifications

- **Microsoft Certified: Azure AI Apps and Agents Developer Associate** (Exam AI-103) – Oct 2026.

Skills

- **Languages & frameworks:** Python (Pandas, NumPy, scikit-learn), SQL, JavaScript, React, FastAPI, HTML/CSS, REST APIs
- **AI & modelling:** LLMs, RAG, agents, prompt engineering, Claude and OpenAI APIs, Hidden Markov Models, Random Forest, K-Means
- **Cloud, tools & business:** AWS (EKS, VPC, ElastiCache), Kubernetes, Docker, Git, PostgreSQL, Excel modelling, PowerPoint, Figma

Leadership & Extra-Curriculars

McKinsey.org Forward Program Oct 2026 – Present

McKinsey & Company (online) *London, UK*

- Structuring ambiguous business problems and turning analysis into clear recommendations.

Course Representative Oct 2025 – Jul 2026

School of Computing & Mathematics, University of Leicester *Leicester, UK*

- Represented 200 students in departmental meetings; raised issues with staff and reported outcomes back.

Retail Volunteer Sep 2025 – Mar 2026

British Heart Foundation *Leicester, UK*

- Served customers, processed donations and handled day-to-day queries.

Languages

- **English:** Native
- **French:** Working Proficiency